İsmail Bilgehan Kazancı

Quality & Process Engineering · SPC · MSA (Gage R&R) · Process Capability · Python

Bursa / Ankara, Türkiye | bilgehankazanci@gmail.com

linkedin.com/in/ismailbilgehankazanci | github.com/DailyDana | dailydana.github.io

SUMMARY

Manufacturing engineering candidate who builds and validates statistical quality tools. Built an open-source SPC and Gage R&R analysis tool whose results match the reference studies published by Minitab and SAS. Open to internships and full-time roles in quality, process and manufacturing engineering.

SKILLS

Quality: Statistical Process Control (SPC), control charts ($\bar{X}$-R, $\bar{X}$-S, I-MR), Western Electric and Nelson rules, process capability (Cp, Cpk, Pp, Ppk), Measurement System Analysis (MSA), Gage R&R (ANOVA, AIAG Average & Range), Type-1 gage study (Cg/Cgk), non-normal capability (percentile, Box-Cox), normality tests, Six Sigma

Manufacturing: Process planning (stock and machining allowance, CNC milling operation sequence), assembly line balancing (Hoffmann, Kilbridge-Wester, ranked positional weight), group technology (Opitz coding), CNC turning, technical drawing

CAD/CAM: SolidWorks, Siemens NX, Mastercam, AutoCAD; GD&T (datums, form, orientation, runout and position tolerances)

Software & data: Python (pandas, NumPy, SciPy, Matplotlib, Streamlit, pytest), Power BI (basic), PowerShell, Git, GitHub Actions

Languages: Turkish (native) · English (professional working proficiency)

PROJECTS

SPC & Process Capability Analyzer | Python, Streamlit 2026
github.com/DailyDana/process-capability-analyzer

- $\bar{X}$-R, $\bar{X}$-S and I-MR control charts; 4 Western Electric and 8 Nelson rules; within (Cp/Cpk) and overall (Pp/Ppk) capability kept distinct; capability from unstable processes is flagged.
- Crossed Gage R&R (ANOVA and AIAG Average & Range): matches Minitab's published results for the AIAG MSA manual example (%GRR 27.86; %Tolerance 22.68; ndc 4) and SAS variance components for the Houf & Burman data to 5 decimals.
- Type-1 gage study (matches the Bosch Booklet 10 examples) and non-normal Cpk; control limits match published values for Montgomery's piston-ring data; 961 automated tests, CI, CSV/Excel/PDF reporting, live demo: spc-analyzer.streamlit.app

Computer-Integrated Manufacturing coursework (İMM 348) | SolidWorks 2026

- GD&T on technical drawings: A-B-C datums, flatness, perpendicularity, parallelism, circular runout and position at MMC.
- Stock sizing and machining allowance for an AISI 1020 part, 3-axis CNC milling operation sequence; fixture assembly drawing with ISO standard parts, bill of materials and assembly sequence plan.

Hoffmann Assembly Line Balancing | Python, Streamlit 2026
github.com/DailyDana/hoffmann-line-balancing

- Hoffmann precedence-matrix method for SALBP-1; cycle-time sweep, CLI and five-language Streamlit UI.
- Found that the original coursework code was first-fit, not Hoffmann, and rewrote it; cross-checked against brute-force and bitmask-DP solvers, 127 tests.

CAD Work | SolidWorks, Siemens NX 2026
github.com/DailyDana/cannon-model

- Naval cannon in SolidWorks with 8 parts, 2 assemblies and full drawings; 19-part Napoleon field gun assembly and assembly drawing in Siemens NX.

EXPERIENCE

Chair of the Audit Board April 2026 – Present
HSD Gazi (HUAWEI Student Developers, Gazi University community) | Ankara

- Leads the community's audit board; designed and built the community website (live: dailydana.github.io/hsd-gazi-web).

IT Systems Assistant (seasonal) June 2023 (1 month)
Doruk Hospitals | Bursa

- Hardware and software installation, on-site and remote user support.
- Network cabling, asset inventory, laptop and hard-disk fault diagnosis and repair.

EDUCATION

B.Sc. Manufacturing Engineering Ankara
Gazi University, Faculty of Technology

- Hands-on CNC, CAD/CAM and technical drawing coursework; introductory CMM session; focus on SPC and quality.

CERTIFICATIONS

- Six Sigma White Belt — The Council for Six Sigma Certification (CSSC), September 2026 (ID: jJSN0zWra1)
- Project and Risk Management — BTK Akademi, March 2026
- SOLIDWORKS: Advanced Engineering Drawings; Project Management Tips — LinkedIn Learning, March 2026
- Explore Data Analysis; Get Started Building with Power BI — Microsoft Learn, 2026